

Ensemble Methods Combination for Motor Imagery Tasks in Brain Computer Interface

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Abstract—The correct recognition of Motor Imagery task in Brain-Computer Interface (BCI) systems has been an important issue in recent studies. In this study, we propose a classification framework based on ensemble methods to handle spectral and spatial EEG signal characteristics. A mixture of two ensemble classifiers has been used for combining multiple information sources. The performance of the proposed classifier has been evaluated on a two-class problem (right and left hand) from the BCI Competition IV dataset 2a. The used features for the training data are the selected features by Mutual information-based Best Individual Feature from the output of the Filter Bank Common Spatial Pattern. The results show that proposed method can reach an accuracy of 90.27% with just 7 features, while other methods have lower accuracy and a higher number of features.

Keywords; EEG; Brain-Computer Interface (BCI); Classification; Ensemble Methods

I. INTRODUCTION

A brain-computer interface (BCI) system, enables disabled people to interact with the environment, without muscles and peripheral nerves involvement, using brain activity Electroencephalogram (EEG) signals [1]. Most BCIs are EEG-based due to its noninvasive property and reasonable recording equipment [2]. A non-muscular pathway is created to transform brain activities into control commands which then are used in external devices like computers, speech synthesizer, assistive appliance, and neural prostheses [3].

Motor imagery based BCIs are one of the multiple types of BCIs. This subfield is based on understandable principles, which makes it very interesting [4]. They are usually designed to transfer two different commands (right and left hand movements). Due to the reverse relation between the accuracy and number of classes, the increase in the number of tasks may lead to misclassification [3].

There are some patterns to obtain the brain activities from the EEG measurements. In recent years, spatial and spectral patterns, as well as extracted features combining from independent sources, have been investigated to boost the performance [1]. Common Spatial Pattern (CSP) is one of the most popular spatial filters in MI-BCI systems which greatly

depends on appropriate selection of frequency bands. According to the EEG signal properties, extracted features from each sub-band filter have a high intersubject variability. To address the mentioned issue, several approaches have been proposed. Sub-band Common Spatial Pattern (SCSP) [5], Filter Bank CSP (FBCSP) [6], Common Spatio-Spectral Pattern (CSSP) [7] and Sparse FBCSP (SFBCSP) [8] are the most effective proposed methods. The FBCSP [9] uses CSP features from nine fixed band pass filters and selects the significant features by Mutual Information-based Best Individual Feature (MIBIF) algorithm. Reported results in this research are evaluated on BCI Competition IV datasets 2a and 2b. Most of the literature in FBCSP are focused on the feature selection section. Sparse dictionary learning method [8], Fisher score, laplacian score [10] and MIBIF are used.

Classification phase is the other important section in the recognition of motor imagery tasks. With this perspective, a new algorithm is introduced in this paper. One of the state-of-the-art studies on ensemble method is done in [1]. Stacked generalization as ensemble method, based on Linear Discriminant Analysis (LDA) classifiers is used for information combination of multiple sources. [11] investigated the feasibility of three ensemble method in EEG classification of mental imagery tasks. Its findings indicated the acceptable performance of these methods. In [12] KF-SVM has been employed for classification. The kernel-Fisher classification accuracy is as high as, or even higher than SVM. AdaBoost, LPBoost, RUSBoost, Bagging and Random Subspaces are used in [13] as different types of ensemble methods to address a three-class motor imagery EEG data. The base learner is feed forward neural network (FFNN). Their findings show that ensemble methods perform better than a single FFNN, except for RUSBoost. Ensemble SVM (ESVM) is the classifier used in [14] to classify noisy EEG signals. ESVM employs several SVM classifiers to tackle the problem of time and subject varying. Training data is divided into separate clusters, then each of them is used to train one single SVM. In the presence of noise, ESVM has a higher accuracy. The base classifier of [15] is LDA. Every LDA is applied on a part of training data that has been generated by bootstrapping. LDA is also the base classifier of [3]. Stacking is used to utilize the ensemble method power to

mix the information of different sources and diminish the EEG signal uncertainty. Results show its appropriate performance.

There are two reasons that make the ensemble method appropriate for EEG classification. First, the EEG dimensionality is usually high and one of the BCI requirements is to train the classifier as soon as possible. Hence, the training set should be small. Second, because of the time-varying feature of EEG signals, it is risky to train just one classifier in order to determine the unknown features [13]. In this study, we propose a new classification method that utilizes the concept of ensemble classification to handle two-class (left and right hand) information sources. Three different bagging models have been combined in level-0 of stacking to investigate the effect of these two ensemble methods mixture. The performance of this method has been assessed on the BCI Competition IV dataset 2a. The results are compared to three prevalent simple classifications including SVM, LDA, and DT, as well as ensemble classifications such as bagging and stacking.

The rest of this paper is organized as follows. Section 2 describes our method in 4 separate signal processing and machine learning parts. Experimental results are presented in section 3 and discussed in section 4.

II. METHOD AND MATERIAL

The structure of the proposed method, consists of 4 signal processing and machine learning stages: band-pass filtering, extracting features by CSP algorithm, selecting features using Mutual Information (MI) algorithm and a new approach for classification. The road map of the mentioned method is shown in Fig. 1.

A. Feature Extraction and Selection

1) Band-Pass Filter

Instead of using a wide filter band, raw EEG signals are filtered by 17 overlapping sub-bands: 4-8Hz, 6-10Hz... 36-40Hz [1].

2) Feature Extraction

Common Spatial Pattern (CSP) is one of the most effective filtering algorithms for motor imagery based BCI systems [8]. Spatial filtering with CSP can be represented by a linear transformation on EEG recorded signal formulating as follows:

$$Z_b = W_b X_b \quad (1)$$

Here $X_b \in R^{N_c \times S}$ denotes the filtered raw signal from b th band-pass filter that can be viewed as (Channel $\times$ measurement samples). The Spatial filter W_b projects the X_b to the brain source signals (Z_b).

The base of CSP algorithm is forming a covariance matrix which discriminates two data classes by maximizing the variance of one class while the other class variance will be minimized. Mathematical overview of the mentioned issue can be shown as the following optimization problem:

$$J(w_b) \triangleq \frac{w_b^T X_{1,b} X_{1,b}^T w_b}{w_b^T X_{2,b} X_{2,b}^T w_b}, \quad (2)$$

where $X_{1,b}$ and $X_{2,b}$ are signals from b th band-pass filter referring to the i th class ($i \in (1, 2)$). After calculating the spatial filter, the features of k th signal trial ($k=1, 2 \dots K$) from the b th filter are given by:

$$v_{b,k} = \log \left(\frac{\text{diag}(Z_{b,k} Z_{b,k}^T)}{\text{trace}[Z_{b,k} Z_{b,k}^T]} \right), \quad (3)$$

where $v_{b,k} \in R^{1 \times 2m}$ has $2m$ features of trial k from b th sub-band. $\text{diag}(\cdot)$ returns the diagonal elements of the square matrix; $\text{tr}[\cdot]$ returns the sum of the square matrix diagonal elements. As already mentioned, there is a total of 9 sub-band filters, so the final feature set is formed as:

$$V = \begin{bmatrix} v_{1,1} & \cdots & v_{1,2m*9} \\ \vdots & \ddots & \vdots \\ v_{K,1} & \cdots & v_{K,2m*9} \end{bmatrix}, \quad (4)$$

where K is the number of all trials.

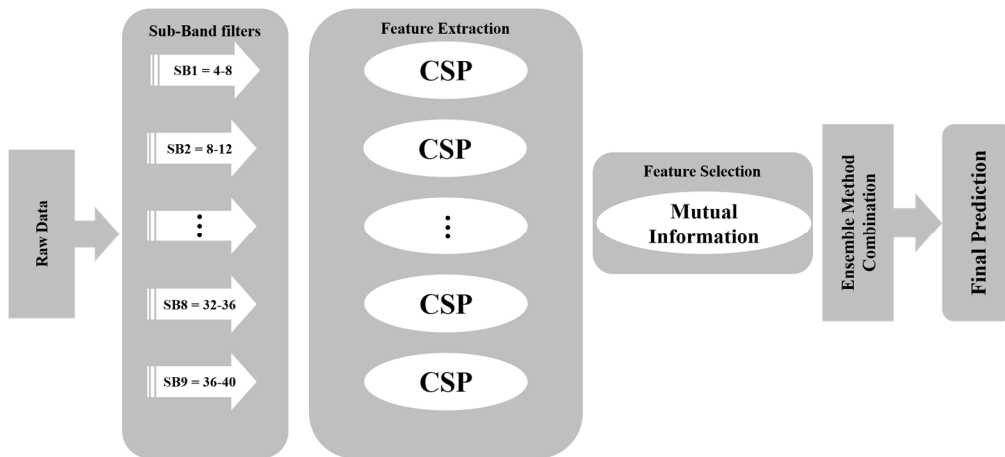


Figure 1 The overview of different processing stages

3) Feature Selection

Some recent researches on filter bank CSP [9] have used the MIBIF algorithm to select the most effective features. The use of the MIBIF algorithm in recent researches on filter bank CSP [3] has been prevalent and it has been applied to our method. After computing the mutual information of each feature and class labels, and sorting them in descending order, this algorithm chooses the most F discriminative features. The number of chosen features (F) has been changed to get the best accuracy during this experiment, Fig. 4 represents the influence of varying F on the accuracy.

B. Classification

1) LDA, SVM, and DT

The classification part plays a significant role in motor imagery-based BCI systems. LDA, Support Vector Machine (SVM) and Decision Tree (DT) are three classifiers which we employed. LDA obtains an acceptable accuracy and its computational requirements are not too high [1]. SVM is another classifier which is effective in high dimensional spaces and in case of nonlinearity, kernel functions can be applied. DT is a non-parametric learning method that is simple to understand and interpret. Its other property is that it can be unstable. Although this property may seem undesirable at first, later on, we will use this quality.

2) Ensemble methods Combination

The combination of several base classifier predictions which is known as ensemble methods usually results in a higher performance. Three families of ensemble classification are Boosting, Bagging, and Stacking. The difference of these groups is in the combining mechanism of their base classifiers' predictions. In bagging, as illustrated in Fig. 2 some random subsets from the main dataset are fed to each base classifier to get a final prediction by aggregating the prediction of each one. Its major goal is to reduce the variance. Stacking has two levels, level-0 consists of several base classifiers and level-1 has only one classifier. For every classifier in level zero, leave-one-out cross-validation has been done. The output of level zero will be considered as level one's features which along with class labels, forms the new training data. The result of the level one's classifier will be the final prediction.

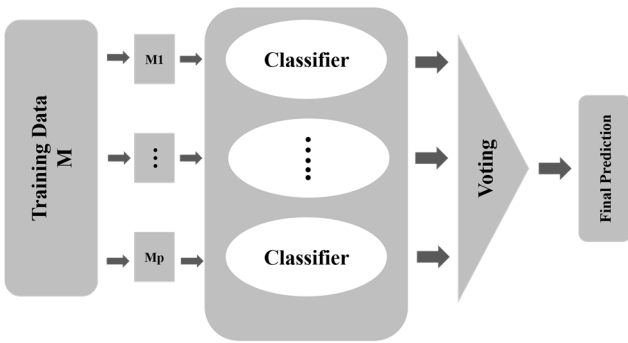


Figure 2 The structure of Bagging

In our method, we want to evaluate the combination effect of stacking and bagging. For this purpose, as illustrated in Fig. 3 we used the concept of bagging in the level-0 of stacking. It means that instead of using three simple classifiers in level-0, we used three meta-learners. For training phase, just like the usual stacking, the input of every meta-learner is the whole training data M . Then each meta-learner consist of k base classifiers, is trained with some random subsets from M ($M_1, M_2, \dots, M_p$). The goal of this part is to predict the training data. The output of each meta-learner (training data prediction) is used as features of level-1 to train it. For testing phase, we used the test data and their corresponding classes were computed at the end of the level-1. Here, for different kinds of meta-learners, we calculated the accuracy.

Spatial and spectral information contained in EEG signal were exploited through this method. The input of level-0 classifiers is features selected by CSP which contain spatial and spectral characteristics of the signal.

III. EXPERIMENTAL RESULTS

The performance of the proposed method is evaluated on BCI Competition IV dataset 2a. Our method is compared to some other classifiers: SVM, LDA, DT, Bagging, and Stacking. The results endorse its better performance.

A. Data Description

BCI Competition IV dataset 2a is EEG Signal from 9 healthy subjects recorded using 22 channels. During recording section, subjects perform one of 4 motor imagery tasks. Class labels of this data, are 1, 2, 3 and 4 for left hand, right hand, foot, and tongue, respectively. Each session contains 288 trials for training phase and 288 trial for the testing phase. The data is extracted from EEG electrodes by windowing recorded signal from 0.5 to 2.5 s after visual cue.

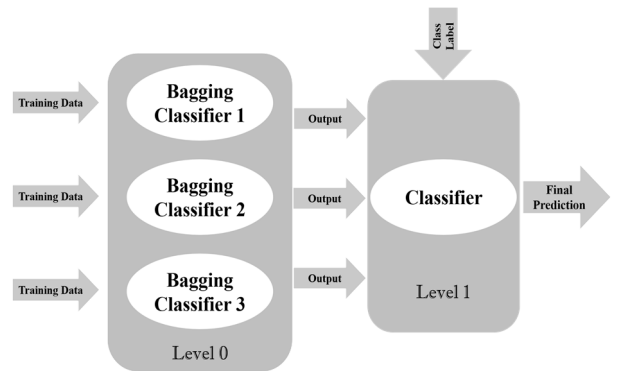


Figure 3 The structure of Proposed Method.

Table 1 The performance of simple classifiers

	F	d		Accuracy %
SVM	15	--	--	70
SVM (Polynomial Kernel)	5	3	--	84.02
SVM (RBF Kernel)	10	--	1.2	90.27
LDA	26	--	--	77.08
DT	25	--	--	62.5

F: Number of Features
D = Polynomial Degree
 σ : RBF parameter

B. Evaluation

As mentioned in section II.A.1), raw EEG signals are filtered by Butterworth (order = 2) to 9 sub-bands: 4-8Hz, 8-12Hz... 36-40 Hz [1]. For feature extraction, CSP algorithm is applied to the output of each sub-band filter and only 6 rows of projected signals are used (half of them from first rows and the rest from last rows) for feature computation. So, $2 \times 3 \times 9$ features are extracted from EEG sub-bands. We changed the value of F (number of features) from 2 to 54 and then the best accuracy is reported.

C. Results

The performance of our method is assessed on two class (left and right hand) of the mentioned dataset. To indicate the higher accuracy of our method, the comparison results are shown in Table 1 and Table 2. There is a possibility that the data is not exactly linear, so we used two kinds of kernel functions (RBF and polynomial) for SVM. RBF SVM obtained the best accuracy with the sigma value of 1.2 and 10 features. As we mentioned before, ensemble methods give higher performance, so we tried a different combination of those three base classifiers.

In Table 2 the results of bagging and stacking are shown. This table has a variety in its number of features. The best classification is the one which has the lower number of feature and higher accuracy. With this logic, in bagging section, DT seemed to be more efficient. In the same number of features ($F=5$), the accuracy of DT, SVM and LDA are 85.41%, 63.19% and 50% respectively.

Table 2 The performance of ensemble classifiers

		F	Accuracy %
Bagging	SVM	19	90.27
	LDA	24	81.94
	DT	5	85.41
Stacking	L0 : SLD , L1 : SVM	26	77.08
	L0 : SLD , L1 : LDA	7	83.33
	L0 : SLD , L1 : DT	26	77.08

F: Number of Features
SLD: SVM – LDA – DT

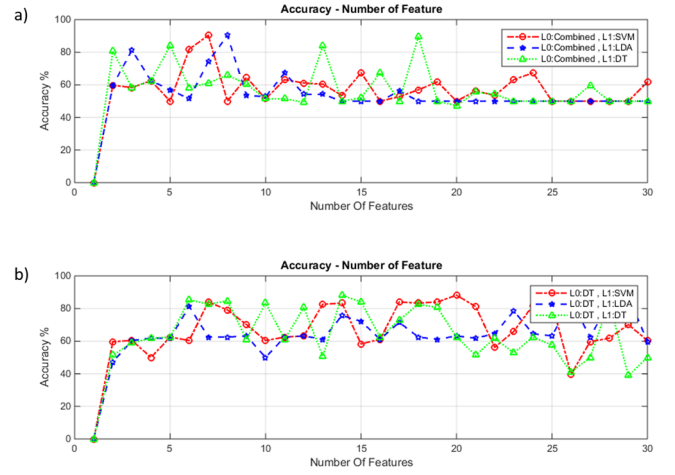


Figure 4 The effect of changing the number of features on accuracy; a) level-0 consists of 3 different bagging classifiers, b) level-0 consists of three DT bagging classifiers

Table 3, represents the results of the proposed method. The first three rows of the table have the same level-0 consisting of three different bagging classifiers and distinct simple base classifiers in the level-1. The second three rows have just DT_{Bag} in their level-0. Results show that the combination in the first row is the best.

IV. DISCUSSION

In this study, we introduced a new classification approach for two-class Motor Imagery BCI systems. We utilized the notion of ensemble methods to elevate the performance of classification. The discriminant features were selected through extracted features by a filter bank and a spatial filtering, using MI algorithm.

Table 3 The performance of the proposed method

Ensemble Combination (Proposed Method)			
	F	Accuracy %	
L0 : SVM_{Bag}, LDA_{Bag}, DT_{Bag} L1 : SVM	7	90.27	
L0 : SVM_{Bag}, LDA_{Bag}, DT_{Bag} L1 : LDA	8	90.27	
L0 : SVM_{Bag}, LDA_{Bag}, DT_{Bag} L1 : DT	6	89.58	
L0 : DT_{Bag}, DT_{Bag}, DT_{Bag} L1 : SVM	20	88.19	
L0 : DT_{Bag}, DT_{Bag}, DT_{Bag} L1 : LDA	29	87.50	
L0 : DT_{Bag}, DT_{Bag}, DT_{Bag} L1 : DT	14	88.19	

As Table 3 reveals, among the three simple classifiers, LDA has the highest accuracy in the cost of a large number of feature. SVM is the next one with acceptable accuracy and fewer features and the use of kernel functions even improved its performance.

In bagging section, although RBF SVM has the best accuracy, its high number of features ($F=19$) is not affordable. Our findings show that DT due to its instability works more effective. In the same number of features, it has the best accuracy among SVM and LDA. In the stacking part, due to the use of three different base classifiers in level-1, we got three different accuracies. The one in which LDA was in the level-1, gave us the maximum accuracy along with a reasonable number of feature.

Among the first three ensemble combinations according to Table 3, the level-1 SVM with a number of feature and accuracy equal to 7 and 90.27% was the best one. The level-1 LDA with just one more feature gives the same accuracy as level-1 SVM. According to the bagging part results, we expected that maybe the use of all DT classifiers in level-0 leads us to a proper reason. So we had three more combinations because of the distinct base classifier in level-1. To show a good aspect of our method, consider the SVM in different sections. In all of them, the accuracy is 90.27%, but the salient difference is their number of features. The proposed method used only 7 features, while the RBF-SVM and bagging-SVM used 10 and 19 features respectively. Finally, besides all the advantages, it is also important to bear in mind that due to the computational cost, the proposed method is more appropriate in off-line applications.

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